CS3 RESEARCH EXPERIENCE FOR TEACHERS (RET)

Colab YOLOv8 Training Tutorial

Training an object detector on your own labeled images in Google Colab

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OVERVIEW

Thirteen steps from a blank notebook to a model

SET UP THE NOTEBOOK

1. Open Colab and press *Upload*
2. Upload the training notebook
3. Press *Connect*
4. Open the *Files* panel

TRAIN ON THE EXAMPLE DATA

5. Add `windows.zip`
6. Learn to run a cell
7. Run the cells up to Step 5
8. Check the predictions

TRAIN ON A ROBOFLOW DATASET

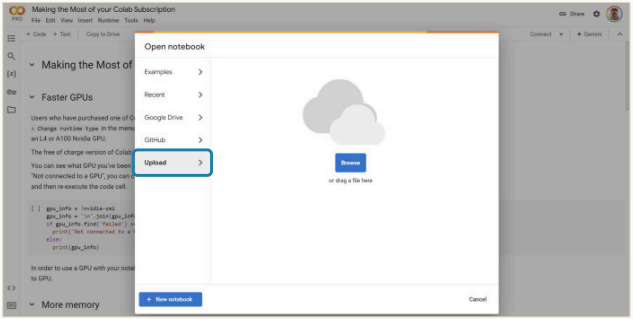
9. Export the dataset from Roboflow
10. Copy the download link
11. Replace the link in Step 5
12. Run the remaining cells
13. Download your model

Training notebook
https://drive.google.com/file/d/1Q2nRfuA020fxSn7IDEHpEVOLurFJAADy/view?usp=sharing

Example dataset
https://drive.google.com/file/d/1tBCSwfhS9NMnNEhIZn6yHwdKYtwrcpFJ/view?usp=sharing

STEP 1

Open Colab and press *Upload*



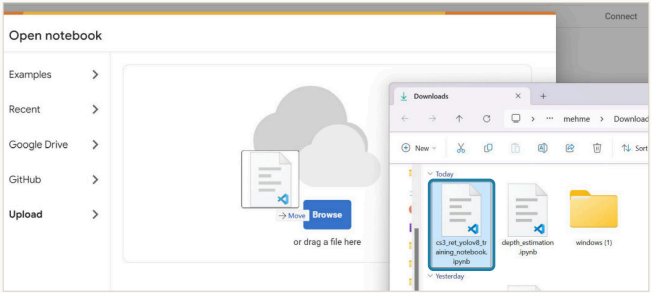
Go to the Colab website:

`https://colab.research.google.com/`

In the *Open notebook* window, press *Upload*.

STEP 2

Upload the training notebook



Download the training notebook from the link below, then drag and drop the file onto the upload area.

Notebook
https://drive.google.com/file/d/1Q2nRfuA020fxSn7IDEHpEVOLurFJAADy/view?usp=sharing

STEP 3

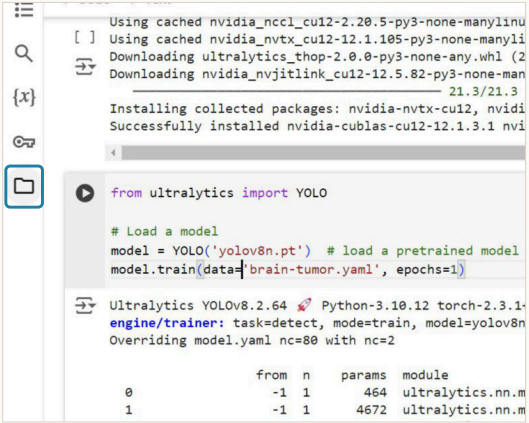
Press *Connect* at the top right



Press *Connect* to start a runtime, the computer in the cloud that runs your notebook.

STEP 4

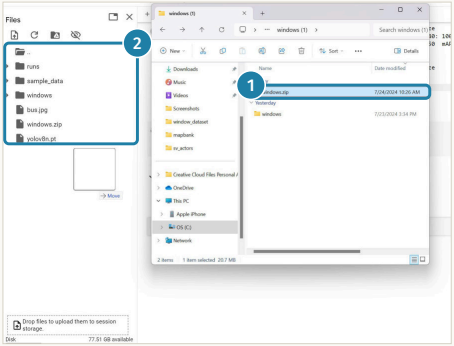
Open the *Files* panel



Click the folder icon in the left sidebar to show the files stored with this session.

STEP 5

Add `windows.zip` to the session files

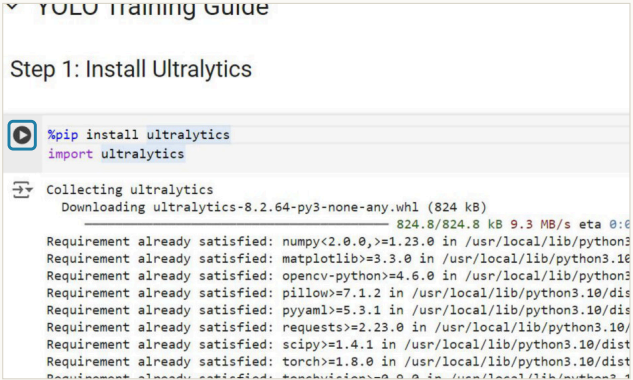


1. Download the example dataset, `windows.zip`, from the link below.
2. Drag it into the *Files* panel on the left. In Colab, add it to the `content` folder.

Example dataset
https://drive.google.com/file/d/1tBCSwfhS9NMnNEhIZn6yHwdKYtwrcpFJ/view?usp=sharing

STEP 6

Run a cell



Click the circular start button at the left of a cell.

Or click inside the cell and press `Shift+Enter`.

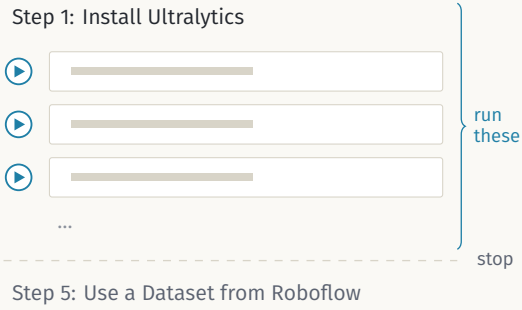
The output appears below the cell.

STEP 7

Run the cells up to the start of Step 5

Run the cells one after another, from the top of the notebook down to the heading *Step 5: Use a Dataset from Roboflow*.

Wait for each cell to finish before you start the next one.



STEP 8

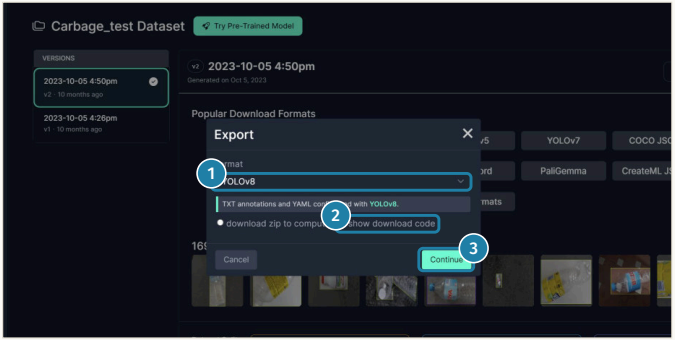
Check the model's predictions



In the *Files* panel, open `runs/detect/train` and find `val_batch0_pred.jpg`. It shows the model's predictions on validation images.

STEP 9

Export your dataset from Roboflow

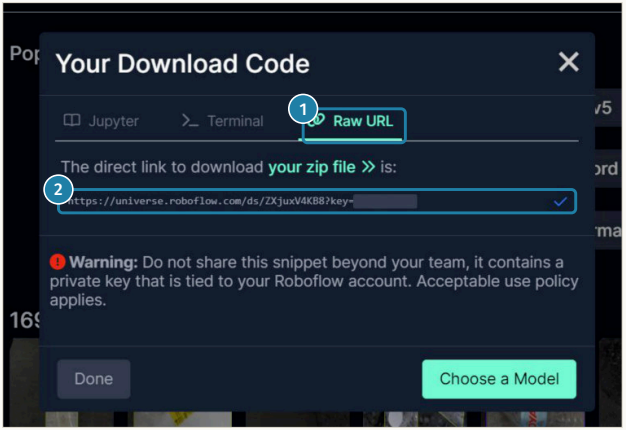


To train on a dataset of your own, open its version in Roboflow and export it.

1. Choose the *YOLOv8* format.
2. Select *show download code*.
3. Press *Continue*.

STEP 10

Copy the download link



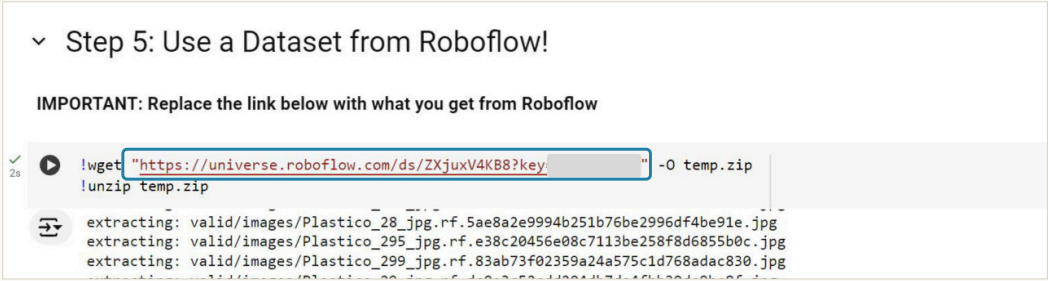
1. Open the *Raw URL* tab.
2. Copy the link.

Keep the link private

As the dialog warns, the link contains a private key tied to your Roboflow account.

STEP 11

Replace the link in Step 5



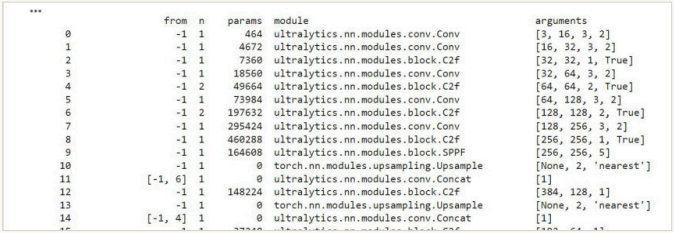
In the notebook, paste your link in place of the one inside the quotation marks after `!wget`, then run the cell.

STEP 12

Run the remaining cells and get your model

```
from ultralytics import YOLO

# Load a model
model = YOLO('yolov8n.pt')  # load a pretrained model (recommended for training)
model.train(data='data.yaml', epochs=10)
```



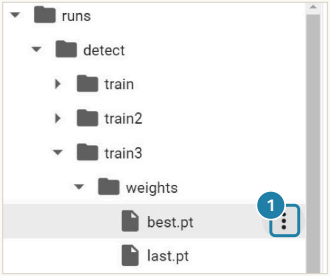
	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]
3	-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
4	-1	2	49664	ultralytics.nn.modules.block.C2f	[64, 64, 2, True]
5	-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
6	-1	2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	1	460288	ultralytics.nn.modules.block.C2f	[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
12	-1	1	148224	ultralytics.nn.modules.block.C2f	[384, 128, 1]
13	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
14	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]

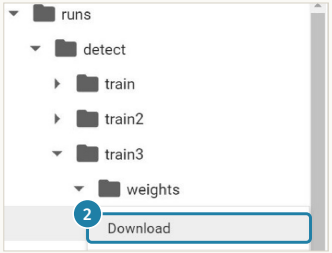
Run the cells that follow Step 5.

The training cell prints the model's layers and then trains on your dataset.

STEP 13

Download your model for future use





Find `best.pt` in the `weights` folder of your training run, under `runs/detect`.

1. Click the three dots next to it.
2. Press *Download*.

> **Remember**
>
> Download `best.pt` so you can use your trained model later.